Valuation of Pine Stands under Uncertainty

by

Charles E. Merritt IV

(Under the direction of Pete Bettinger)

Abstract

Stand-level forest planning is commonly conducted with deterministic growth and yield models that assume a single, idealized trajectory of stand development. However, forest stands evolve under substantial uncertainty arising from stochastic growth variation and disturbance events such as pests, wildfire, and wind damage. This thesis develops a modular stand simulation framework that augments an established deterministic loblolly pine (*Pinus taeda*) growth model with stochastic process noise and probabilistic disturbance regimes, and uses Monte Carlo simulation to quantify how uncertainty alters economic valuation.

Rather than learning management policies through reinforcement learning, we evaluate a set of predefined, interpretable management prescriptions under both deterministic and stochastic dynamics. For each prescription and risk regime, repeated simulations generate a distribution of discounted cash flows and corresponding Net Present Value (NPV), with Land Expectation Value (LEV) reported when appropriate to enable comparison with classical forestry benchmarks. Outcomes are summarized using measures of central tendency and dispersion as well as downside risk metrics, including Value at Risk (VaR), Conditional Value at Risk (CVaR), and the probability of negative NPV.

Results show that deterministic valuation can systematically mischaracterize stand value under uncertainty, particularly when low-probability, high-severity disturbances are present.

Across increasing disturbance frequencies and severities, the stochastic valuation distributions exhibit substantial tail risk and increased variability relative to deterministic baselines, and financially preferred rotation ages and prescriptions may shift accordingly. Collectively, these findings demonstrate the importance of distributional valuation in long-horizon timber management and provide a practical framework for comparing prescription robustness under stochastic disturbance regimes.

Index words: Forest Value, Markov Decision Processes, Forest Growth & Yield Modeling, Simulation, Digital Forestry, Theses (academic)

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B.A., The University of Georgia, 2024

A Thesis Submitted to the Graduate Faculty
of The University of Georgia in Partial Fulfillment
of the
Requirements for the Degree

Master of Science

Athens, Georgia

2026

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July 2026

Dedication

I dedicate this work to my mother, whose unwavering support and guidance have shaped me into the person I am today. Your belief in me has always been my greatest strength. To my siblings, Collier, Caiden, and Annalynn, your eyes inspire me daily to set a worthy example and continuously motivate me to become a better version of myself. Thank you all for keeping me grounded and focused on what truly matters.

Acknowledgments

This work was supported by the U.S. Department of Agriculture, Forest Service, Southern Region, Federal Grant Number 22-CS-11080100-232 (Talladega Division LiDAR Project). This work was also supported by the intramural research program of the U.S. Department of Agriculture, National Institute of Food and Agriculture, Sustainable Agricultural Systems program, PERSEUS grant, #2023-68012-38992. The findings and conclusions in this work have not been formally disseminated by the U. S. Department of Agriculture and should not be construed to represent any agency determination or policy. The U.S. Government is authorized to reproduce and distribute reprints for governmental purposes notwithstanding any copyright annotation therein.

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Chapter 1

Introduction

1.0.1 Background and Context

Forest management decisions often span long time horizons, necessitating methods that anticipate future conditions and trade-offs. Yet most current stand-level planning models are deterministic, assuming a single, idealized trajectory of forest development. These models typically use optimization techniques to identify regimes that maximize a static objective (or set of objectives), most often timber volume or forest structure goals, though stands may also be managed for carbon sequestration, wildlife diversity, or recreation. In doing so, they often overlook a critical reality: forest stands are not deterministic environments.

Rather, forest stands are subject to stochastic and dynamic processes. Growth trajectories are influenced not only by competition and ecological succession, but also by random disturbances such as wildfires, windthrow, ice storms, and pest outbreaks. These events are often abrupt, severe, and difficult to predict. Their effects propagate over time, altering successional pathways and disrupting static management plans. Despite this, such disturbances are often omitted from standard planning methods, which focus on idealized scenarios rather than risk-resilient strategies. Furthermore, disturbance events beyond a threshold of severity substantially increase the likelihood of sustained disturbances. For example, woody debris caused by wind damage increases forest fuel loads and subsequently the likelihood of wildfire while also increasing the susceptibility of trees to infestation from pests such as pine beetles due to their weakened state.

These random disturbance events are projected to become more frequent and severe. Recent history bears this out. For example, Georgia Forestry Commission (2024) reported

that Hurricane Helene caused $1.28 billion in damage to Georgia's forests, and affected up to 8.9 million acres of forest land in Georgia alone. Other events, such as the 2007 Georgia Bay Fire and the 2011 Honey Prairie Fire, burned 560,000 acres and 300,000 acres respectively. These events illustrate the vulnerability of forest systems to random, high-impact events in the Southern United States.

This growing uncertainty calls for a new modeling paradigm, one that explicitly represents stochastic risks and adapts to variable conditions over time. Rather than optimizing for a single, fixed trajectory, we propose a simulation-based framework that captures a range of plausible forest futures. While no simulator can perfectly reproduce the complexity of forest growth and disturbance, our intent is not to create a definitive predictive model for growth *and* stochastic disturbances in timber stands. Instead, the simulation environment serves as a modular testbed for evaluating management strategies and their effects on stand value under uncertainty. Its design allows for replacement or extension of the growth or disturbance models with site-specific or more sophisticated alternatives. By augmenting a deterministic growth & yield simulator with stochastic noise and disturbance events, we construct a flexible modeling environment in which multiple stochastic risk scenarios can be explored and evaluated against non-stochastic baselines.

To structure the stochastic elements in this environment, we model stand development as a probabilistic state transition process. At each annual timestep, stand attributes evolve according to established growth and yield equations, with additional variability introduced through stochastic growth noise and disturbance events. This formulation allows us to generate multiple plausible realizations of stand development under identical management prescriptions.

Rather than solving for an optimal policy through reinforcement learning, this study evaluates predefined, rule-based management prescriptions under both deterministic and stochastic dynamics. The objective is not to learn an adaptive control strategy, but to quantify how uncertainty in growth and disturbance processes alters economic valuation out-

comes. By holding management prescriptions constant and varying the disturbance regime, we isolate the effect of stochastic risk on projected stand value.

Risk is evaluated using distributional financial metrics rather than single point estimates. For each prescription and disturbance regime, Monte Carlo simulation produces a distribution of discounted cash flows and corresponding Net Present Value (NPV). These distributions are summarized using measures of central tendency and dispersion, as well as downside risk metrics such as Value at Risk (VaR) and Conditional Value at Risk (CVaR). This approach allows comparison not only of expected returns but also of resilience to rare but severe disturbance events.

By reframing stand valuation as a distribution rather than a deterministic forecast, this framework provides a more transparent view of the economic consequences of uncertainty. In doing so, it bridges classical deterministic growth and yield modeling with modern risk analysis, while remaining grounded in established forestry finance principles such as NPV and LEV.

This thesis addresses several gaps:

- The ubiquitous tendency to sidestep uncertainty in deterministic optimization models which leave out many probabilistic events that are relevant to growth but complicated to model as described by Stevens (1986).

- Limited use of distributional valuation metrics in stand-level planning, where deterministic point estimates often mask downside risk.

- The missing link between tried and true empirical stand growth models (Harrison et al. 1996) and stochastic models that include both chaotic disturbances and growth factors.

We propose a framework that embraces stochasticity from the outset by embedding probabilistic disturbance processes within an established growth and yield model and evaluating heuristic "rule of thumb" management prescriptions through Monte Carlo simulation. This

approach enables direct comparison between deterministic and stochastic valuation outcomes under varying disturbance regimes.

This thesis investigates how probabilistic disturbance modeling can be integrated with established growth and yield models to evaluate forest stand value under uncertainty.

Stochastic vs. Deterministic Planning

How does deterministic stand valuation differ from stochastic valuation under identical management prescriptions?

This question evaluates the extent to which conventional deterministic planning methods, such as fixed-rotation management optimized via LEV, diverge from valuation results obtained when stochastic disturbances and growth variability are explicitly modeled. Rather than comparing learned policies to deterministic schedules, we hold management prescriptions constant and examine how introducing stochastic disturbance processes alters expected value, variability, and downside risk.

How large is the bias introduced by ignoring disturbance risk and process uncertainty?

Deterministic stand models implicitly assume a single, idealized development trajectory. In contrast, stochastic Monte Carlo simulation generates a distribution of possible futures. This question assesses the difference between deterministic point estimates of value and the mean, median, and tail outcomes of the corresponding stochastic distribution. Particular attention is given to downside risk, including the probability of substantial loss under elevated disturbance regimes.

Do changes in disturbance frequency and severity alter financially optimal rotation ages and prescription performance?

Classical optimal rotation theory identifies a single rotation age T^* under deterministic assumptions. Under stochastic disturbance regimes, however, realized rotation values are

distributed rather than fixed. This question investigates how increasing disturbance frequency and severity shift the distribution of economically optimal rotation ages and whether prescriptions derived under deterministic assumptions remain robust when evaluated across stochastic realizations. In particular, we examine whether higher disturbance risk systematically favors shorter rotations or alternative rule-based prescriptions.

Simulation and Valuation Framework

How can a stochastic stand simulator be structured to evaluate management prescriptions under uncertainty?

We model stand dynamics as a stochastic state transition process governed by mechanistic growth equations and probabilistic disturbance events. Management actions are specified through interpretable, rule-based prescriptions such as fixed rotation ages, thinning thresholds, and salvage responses. Rather than solving for an optimal policy through reinforcement learning, we simulate stand trajectories under each prescription across many Monte Carlo realizations to estimate the distribution of financial outcomes.

Which valuation and risk measures are appropriate for comparing deterministic and stochastic outcomes?

For each simulated trajectory, we compute discounted cash flows to obtain Net Present Value (NPV), and when appropriate, Land Expectation Value (LEV). Across Monte Carlo realizations, we summarize results using measures of central tendency and dispersion, as well as downside risk metrics such as Value at Risk (VaR) and Conditional Value at Risk (CVaR). These measures allow comparison not only of expected returns but also of resilience to catastrophic loss.

Together, these questions clarify how incorporating stochastic disturbance processes changes economic valuation and prescription robustness in stand-level forest management.

Assumptions

Every model has underlying assumptions, and it's worth discussing them early and directly. First and foremost, while the methods presented are general, almost all of this work was conducted with managed, even-aged, loblolly pine (*Pinus taeda*) plantations in mind. This simplifies the effort expended on factors such as species composition and competition. As such, our tree growth dynamics are modeled using region-specific equations provided by the Plantation Management Research Cooperative (PMRC). For the complete formulation of the deterministic growth model, see Harrison et al. (1996).

All stochastic elements (e.g., disturbance probabilities) are stationary within a scenario. The distribution parameters are chosen at the beginning and held fixed through the episode, which is synonymous with the episode of the MDP. Discount rates, and management costs are also stationary and scenario-defined, they don't evolve stochastically within a simulation. Stationarity is an important mathematical assumption for Markov processes, and the goal of this work is not to model the disturbance probabilities or economic factors themselves, but to understand the effect of these stochastic elements on stand value. For each of these values, we use the most recent estimates available from the literature.

Disturbances may have lagged effects (e.g., pest outbreaks causing gradual decline), which are captured through recovery functions but assumed parametric and consistent. In this work, disturbances do not alter future disturbance probabilities (e.g., one fire doesn't reduce or increase the chance of another, except indirectly through changed stand state). When disturbances occur, a salvage fraction of 50% is assumed available as revenue. We don't model harvest scheduling constraints, road access, policy restrictions, or multi-stand interactions, as this is framed as a single stand modeling problem for tractability.

Finally, while the stand transition process is Markovian in structure, this assumption serves as a modeling abstraction rather than a literal claim about ecological memorylessness. The state vector is constructed to capture sufficient information for future growth and disturbance response within the scope of this study. It is very likely that the true ecolog-

ical process is not Markovian, but this is a common assumption in MDPs and is a useful abstraction for the purposes of this work.

1.1 Thesis Statement

This thesis finds a significant gap between the projected stand value at the end of a rotation from deterministic and stochastic simulations, even under modest estimates of growth variability, disturbance likelihood,and identical management regimes.

Chapter 2

Related Works

2.1 Theoretical Framework

This thesis is grounded in the theory of Markov Decision Processes (MDPs). MDPs formalize the problem of sequential decision-making under uncertainty, offering a mathematical structure in which agents take actions, receive rewards, and observe the consequences over time. Markov processes are subject to the Markov assumption, which states that a Markovian process is one in which the next state is entirely determined by the current state and action, giving a definitive structure to the environment within which a learning agent can be embedded. Any Markov Decision Process (MDP) can be defined by the 5-tuple:

$$\{\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma\}$$

where:

- $\mathcal{S}$ is the set of possible states,
- $\mathcal{A}$ is the set of possible actions,
- $\mathcal{P}$ is the state transition probability function $\mathcal{P}(s' \mid s, a)$,
- $\mathcal{R}$ is the reward function $\mathcal{R}(s, a, s')$,
- $\gamma \in [0, 1]$ is the discount factor.

Furthermore, for finite-horizon problems, we additionally specify a horizon T, the number of decision steps left until termination. This is an important parameter for MDPs, as the

amount of steps left in your planning horizon affects the overall policy strongly. Infinite-horizon MDPs, which are not the focus of this work, always converge to a unique policy. In forestry, we are interested in the fixed rotation length befitting southen pine plantations, and so work with a fixed horizon roughly comparable to this timeframe.

Forest management, particularly at the stand level, can be naturally modeled within this structure. At each timestep, a decision-maker observes a forest stand's condition (e.g., trees per acre, basal area, disturbance status, etc.) and can decide upon a course of action. Rewards are typically related to timber financial value, ecosystem services, or long-term economic returns and are accrued based on both the current state and the chosen action.

Generic MDPs assume full observability of the state of the system. In the context of forest management, this might include full knowledge of the condition of the stand, the exact probabilities of disturbance, etc. However, in real-world forest applications, such perfect information is rarely available. For example, managers often do not know the precise likelihood of a fire occurring in the coming year, but may infer increased risk based on weather patterns or historical fire activity. Furthermore, to satisfy the Markov assumption, we need to construct a state vector that is "Markov-complete". For a state to be considered Markov-complete, the state's representation must contain *all* of the information needed for the transition and reward functions of the environment to depend *only* on the current state and action pair. This is a significant challenge if one considers the temporal and dynamic complexity of even a single forest stand. Many factors will introduce lingering effects over many time steps, and we would need to include these in our state vector for the Markov assumption to hold. While we stop short of implementing partial observability in this work, it's worth acknowledging as an avenue for future directions. In a partially observable MDP (POMDP), the agent does not observe the true state directly, but instead receives an observation $o \in O$ generated according to an observation function $\mathcal{Z}$. In this formulation, the agent maintains a belief state that is a probability distribution over the possible underlying states, and uses this to inform its actions. This framework would more closely reflect the

reality of the conditions under which forest managers really operate, but comes with a unique set of challenges that are outside of the scope of the current work.

2.1.1 Reinforcement Learning as a Guiding Principle for Stochastic Modeling

Reinforcement learning (RL) is a family of methods for solving MDPs through interaction with the environment. Unlike other approaches that require explicit models of transition and reward functions, this makes RL well-suited for settings where the environment is complex or partially known, and where traditional optimization techniques struggle to scale or adapt.

RL has been successfully applied to a wide range of problems, including robotics, game playing, and resource management. In the context of forestry, RL has been used to address wildfire mitigation (Altamimi et al. 2022), approximate optimal solutions to the harvest scheduling problem (Ji et al. 2024), and more (Malo et al. 2021), (Tahvonen et al. 2022). TK: Add more RL citations and probably cite Dr. Bettinger & Dr. da Silva.

2.2 Current State of Knowledge

2.2.1 Net Present Value and Land Expectation Value

The standard approach to optimizing the set of management decisions (equivalently the "management regime" or the "policy" in an RL context) is to first determine the value of a single rotation for a stand and a given rotation age based on Net Present Value (NPV).

The net present value (NPV) of harvesting a forest stand at rotation age T can be expressed as:

$$NPV(T) = \sum_{t=0}^{T} \frac{R_t - C_t}{(1+r)^t}$$

where:

- R_t is the revenue received at time t (e.g., from thinning or final harvest),

- C_t is the cost incurred at time t (e.g., planting, management, or treatment costs),
- r is the discount rate,
- T is the final rotation age.

Then, we expand on the definition of NPV to account for an infinite series of identical, even-aged rotations, starting from bare land. This is known as the Land Expectation Value (LEV). This formulation assumes that the same stand is regenerated and harvested infinitely many times on the same land. It can be expressed as:

$$LEV(T) = \frac{NPV(T)}{(1+r)^T - 1}$$

where:

- $NPV(T)$ is the net present value of one rotation of length T,
- r is the discount rate,
- T is the rotation age.

The optimal rotation age T^* is defined as:

$$T^* = \arg\max_T LEV(T).$$

Once T^* has been defined for a stand, it is constant for the foreseeable future as long as business proceeds "as usual". Finding T^* is then a matter of some importance, it defines our "infinite and identical rotations" for the LEV. The standard method for establishing the optimal rotation age T^* is to use the LEV to compare multiple rotation ages until you converge on the rotation age that returns the highest value. There are other methods for determining the rotation age, for instance one could use the biological method. In that case, when the mean annual growth rate is maximized, it is time to cut the stand. Both methods yield viable choices for the optimal rotation age, and there are many other viable methods.

For a more detailed examination of methods relating to the assignment of T^* see Bettinger and Li (2009).

This method for calculating the value of the stand holds us to the assumption of "infinite and identical rotations at the optimal rotation age" in order to calculate the LEV. This assumption is extremely limiting, and in fact implicitly avoids uncertain outcomes. Nevertheless it remains an effective, easy, and robust way to calculate the economic value of a stand.

2.2.2 Stochastic Forest Disturbance Modeling

Accurate representation of forest disturbances is critical to keep our simulation as close as possible to the real dynamics of forest growth. To this end, we rely on the following references.

- "A hierarchical simulation-through-optimization approach to forest disturbance modeling" (Campbell et al. 2007) This work offers insights into a Markov approach for forest-level modeling. They very simply alter the standard harvest scheduling approach by treating disturbances as if they were another type of harvesting activity with no revenue or cost. In this way, disturbed stands are implicitly seen as having no value. This method is general enough to allow them to "model any stand-replacing disturbance..." regardless of age. In calculating the likelihood of these disturbances for their spatial model, firstly they retrieved historical data for their study area that had similar rates of disturbances and harvest. Then, they discretize the likelihood of disturbance into decade-long bins, the same time-step used in their Markov model. Campbell & Dewhurst offer a comprehensive methodology for defining a spatial Markov chain for forest-level modeling, especially with regards to getting an implementation of disturbances themselves that is not too complicated.

- "Optimal Harvest Rates Considering the Risk of Forest Fire" (Stevens 1986) This thesis deals with a single risk type and offers some guidance on the ways in which people have

tried to solve these problems historically. Specifically it compares a traditional "even-flow" model with a dynamic programming model to compare how fire-rate impacts harvest volumes.

- "Dealing with Non-linearity and Uncertainty in Forest Management" (Messier et al. 2016)
- "Stochastic or deterministic single-tree models: is there any difference in growth predictions?" (Fortin et al. 2012) Critically, this paper shows that there are differences between stochastic and deterministic simulations due to interactions between "non-linear mechanisms underlying forest growth."
- "Forest Growth and Yield Modeling" (Weiskittel 2011) This textbook provides a comprehensive view of several major techniques for forest models, but also a sizable section on stand-level modeling as well.

2.2.3 Stand-Level Optimization Practices

- "Yield Prediction and Growth Projection for Site-Prepared Loblolly Pine Plantations in the Carolinas, Georgia, Alabama and Florida" (Harrison et al. 1996) This work provides us with robust yield curves and growth equations that provide us with a basis for growth in our MDP model.
- "A density-dependent stand-level optimization approach for deriving management prescriptions for interior northwest (USA) landscapes" (Bettinger, Graetz, et al. 2005)

2.2.4 Markov Decision Processes in Forest Modeling

- "Managing Forests for Carbon and Timber: A Markov Decision Model of Uneven-aged Forest Management With Risk" (Johnston et al. 2017)
- "A Graph-based Markov Decision Process approach for managing forests under risk of wind damage" (Forsell et al. 2007)

This paper explores the use of a MDP in a forest-level management problem subject to wind damages.

2.2.5 Reinforcement Learning in Forest Management

Reinforcement learning builds on the MDP framework by providing computational methods to learn optimal or near-optimal policies from repeated experiences, rather than solving equations for a known model. Early work treating forest management as an RL problem has shown promising results. Malo et al. (2021) applied RL to a high-dimensional forestry scenario: a mixed-species, age-structured stand management problem with stochastic growth and prices. They report that a deep RL agent (incorporating value function approximation) could solve this problem without the simplifying assumptions needed in prior models. Interestingly, their results revealed that conventional low-dimensional models (e.g., a single-age optimal rotation model or a matrix model) either lost a large fraction of potential economic value or misestimated optimal harvest timing, whereas the RL-derived policy captured more value and provided a more accurate assessment of risk. In essence, RL enabled optimization in an environment with far more state variables (tree size distributions, species mix, etc.) than traditional models handle, indicating the potential for RL to be applied to forestry more broadly.

Another concrete example is the use of Q-learning for forest planning. Ji et al. (2024) developed a 60-year forest harvest scheduling model and employed Q-learning (a model-free RL algorithm) to identify harvest policies that balanced economic and ecological goals. Their study, focused on South Korean national forests, defined multiple scenarios with different management priorities (timber production vs. carbon storage vs. other ecosystem services). The RL agent learned policies under each scenario and was able to improve upon baseline management plans. One scenario that prioritized carbon storage not only sequestered more CO_2 but also yielded the highest economic value in the long term, a win-win outcome. The authors conclude that "reinforcement learning is a powerful tool for developing long-term

forest management strategies that address multiple objectives". This exemplifies how RL can handle multi-objective optimization, a common need in natural resource management where trade-offs (timber vs carbon vs biodiversity) must be navigated.

In summary, prior applications of RL in natural resource management demonstrate: (1) the ability to handle high-dimensional or continuous state spaces (e.g. detailed forest stands), (2) accommodation of stochastic dynamics (growth variability, random fires), and (3) tackling multiobjective outcomes by encoding them in reward functions. These successes in the literature provide a strong rationale for using RL as a core component of this project.

- "Timber Harvest Planning Using Reinforcement Learning: A Feasibility Study" (Ji et al. 2024)
- "Reinforcement learning in optimizing forest management" (Malo et al. 2021)
- "Optimizing high-dimensional stochastic forestry via reinforcement learning" (Tahvonen et al. 2022)
- "Large-Scale Wildfire Mitigation Through Deep Reinforcement Learning" (Altamimi et al. 2022)
- "A multi-objective Monte Carlo tree search for forest harvest scheduling" (Neto et al. 2020)
- "Adaptive economic and ecological forest management under risk" (Buongiorno et al. 2015)

2.3 Research Gaps

Chapter 3

Methods

3.1 Research Design

This study evaluates how deterministic and stochastic disturbance dynamics influence the economic valuation of a single forest stand under predefined, rule-based management prescriptions. Rather than learning an optimal policy through reinforcement learning, we specify a set of interpretable prescriptions ex ante and compare their financial performance under deterministic and stochastic growth scenarios.

Management decisions are applied at the stand level and include treatments such as pre-commercial thinning, commercial harvest, and fertilization. Thinning is modeled as a proportion of standing volume removed (assumed thinning from below), while fertilization is applied at a fixed regional rate of 169 lbs/acre (Maggard et al. 2022). Each treatment carries a predefined cost, and harvested timber generates revenue based on product class stumpage prices obtained from TimberMart-South data. Treatment costs are assumed stationary within each scenario, while timber product prices fluctuate randomly between higher, normal, and lower states due to unpredictable market shifts.

The simulation framework models stand dynamics through annual state transitions. The stand state at time t, denoted $\mathcal{S}_t$, is represented as a vector of stand-level attributes including standing volume, trees per acre, mean height, and basal area. The stand is assumed to be even-aged and single-species (loblolly pine, *Pinus taeda*), consistent with plantation conditions. State transitions occur in yearly increments.

Although the transition structure is Markovian in form, this is used as a modeling abstraction rather than a claim of strict ecological memorylessness. The state vector is constructed

to contain sufficient information for projecting future growth and disturbance effects within the scope of this study.

The simulation environment consists of three primary components: a growth module, a disturbance module, and an economics module.

3.1.1 Management Prescriptions

Management decisions are implemented through predefined, rule-based prescriptions rather than optimized control policies. Each prescription specifies treatment timing and intensity as a deterministic function of stand state variables. The prescriptions evaluated in this study include:

TK: Not all implemented**Fixed Rotation (FR-T)**

The stand is harvested at a predetermined rotation age T. No intermediate treatments are applied unless explicitly specified. We evaluate rotation ages of 25, 30, 35, and 40 years to represent a plausible range for loblolly pine plantations. **Biological Rotation (MAI Rule)**

The stand is harvested when mean annual increment (MAI) reaches its maximum. This rule represents the classical biological rotation criterion. **Basal Area Threshold Thinning (BAT)**

A thinning from below is triggered when basal area exceeds a threshold BA^*. A fixed proportion τ of merchantable volume is removed at that time. Harvest occurs at a specified rotation age T. **Salvage-Responsive Rotation (SR)**

If a catastrophic disturbance reduces standing volume below a specified fraction of pre-disturbance volume, the stand is immediately harvested and regenerated. Otherwise, harvest follows a fixed rotation age. **Fertilization-Augmented Rotation (FA)**

Fertilization is applied once at a predefined stand age (e.g., age 10), increasing subsequent growth according to the growth module parameters. Final harvest follows a fixed rotation age.

Each prescription is applied identically across deterministic and stochastic simulations. This ensures that differences in financial outcomes arise from disturbance and growth variability rather than adaptive changes in management.

3.1.2 Growth Module

The growth module projects ordinary stand development using equations from Harrison et al. (1996). Annual growth increments follow a deterministic function of the form

$$\Delta s = f(s_t, a_t; \theta) \tag{3.1}$$

where s_t is the current stand state and θ denotes parameters derived from the PMRC yield equations. Optional stochastic noise may be added to represent process uncertainty in growth.

The growth module is validated by comparing simulated trajectories to empirical growth patterns reported for managed loblolly pine plantations. Separating growth from disturbance dynamics allows clear interpretation of baseline stand development prior to stochastic modification.

3.1.3 Disturbance Module

Disturbances are modeled as stochastic processes layered onto baseline growth. Two disturbance classes are considered: chronic and catastrophic. Each class is associated with scenario-specific annual probabilities and severity ranges.

Chronic disturbances cause moderate proportional volume losses (5–15%), while catastrophic disturbances result in severe losses (50–90%) followed by multi-year recovery dynamics. Recovery effects are modeled through parametric post-disturbance growth envelopes.

Disturbance probabilities are stationary within a scenario. Three risk regimes (low, medium, high) are defined to assess how increasing disturbance frequency influences expected

value and downside risk. Monte Carlo simulation is used to generate multiple realizations of stand trajectories under each regime.

To ensure statistical rigor, each prescription–scenario combination is evaluated across many independent simulations. Financial outcomes are summarized using sample means, variances, and tail-risk measures. When comparing prescriptions, common random number streams are used to reduce variance in comparative estimates.

3.1.4 Economics and Cashflow Modeling

The economics module converts simulated harvest and treatment decisions into annual cashflows. Revenues are calculated from harvested product classes (pulpwood, chip-and-saw, sawtimber) using stumpage prices that switch between high, average, and low state of demand. Treatment costs are assigned according to regional estimates (Maggard et al. 2022).

For each year t, net cashflow is defined as

$$CF_t = \text{Revenue}_t - \text{Cost}_t.$$

If a catastrophic disturbance occurs, a salvage fraction of 50% of damaged merchantable volume is assumed recoverable as revenue.

Financial performance over a rotation of length T is evaluated using Net Present Value (NPV):

$$NPV = \sum_{t=0}^{T} \frac{CF_t}{(1+r)^t},$$

where r is the discount rate. When appropriate, Land Expectation Value (LEV) is derived from the rotation-level NPV to enable comparison with classical deterministic forestry benchmarks.

3.1.5 Evaluation Metrics

For each management prescription and disturbance regime, Monte Carlo simulation produces a distribution of NPV outcomes. Results are summarized using:

- Mean and median NPV,
- Standard deviation,
- Value at Risk (VaR),
- Conditional Value at Risk (CVaR),
- Probability of negative NPV or severe loss.

Deterministic valuation is obtained by running the same prescription with disturbance processes disabled. The difference between deterministic NPV and the Monte Carlo mean provides an estimate of bias introduced by ignoring stochastic risk.

This design allows direct comparison between deterministic planning assumptions and stochastic valuation outcomes, highlighting how disturbance frequency and severity alter expected returns and downside exposure.

Chapter 4

Results

For analysis of the stochastic MDP model, we consider two baselines: the deterministic case where there is no stochastic risk whatsoever, which is equivalent to running the deterministic model (Harrison et al. 1996) directly, and another where we include process risk but disregard the catastrophic risk, which is equivalent to running the aforementioned deterministic model with the addition of noisy growth functions.

We observe that increasing the rate of catastrophic events leads to a dramatic reduction in the quality of the stand over time compared to the deterministic baseline. Basal area, and dominant height are substantially reduced from the deterministic case, while trees per acre stays near the level the stand was established at due to the effects of recruitment of new trees to the lowest size class following suit with the work of (Buongiorno et al. 2015). This in effect demonstrates that when stands are under conditions of heightened risk from disturbance, they stay at or above a level of competition established at planting while overall productivity from BA and HD growth falls to a level characterized by stands under 5 years of age.

As expected, the deterministic baseline consistently overestimates the value of the stand over time, while the stochastic baseline shows a more realistic decline in value as the frequency of catastrophic events increases.

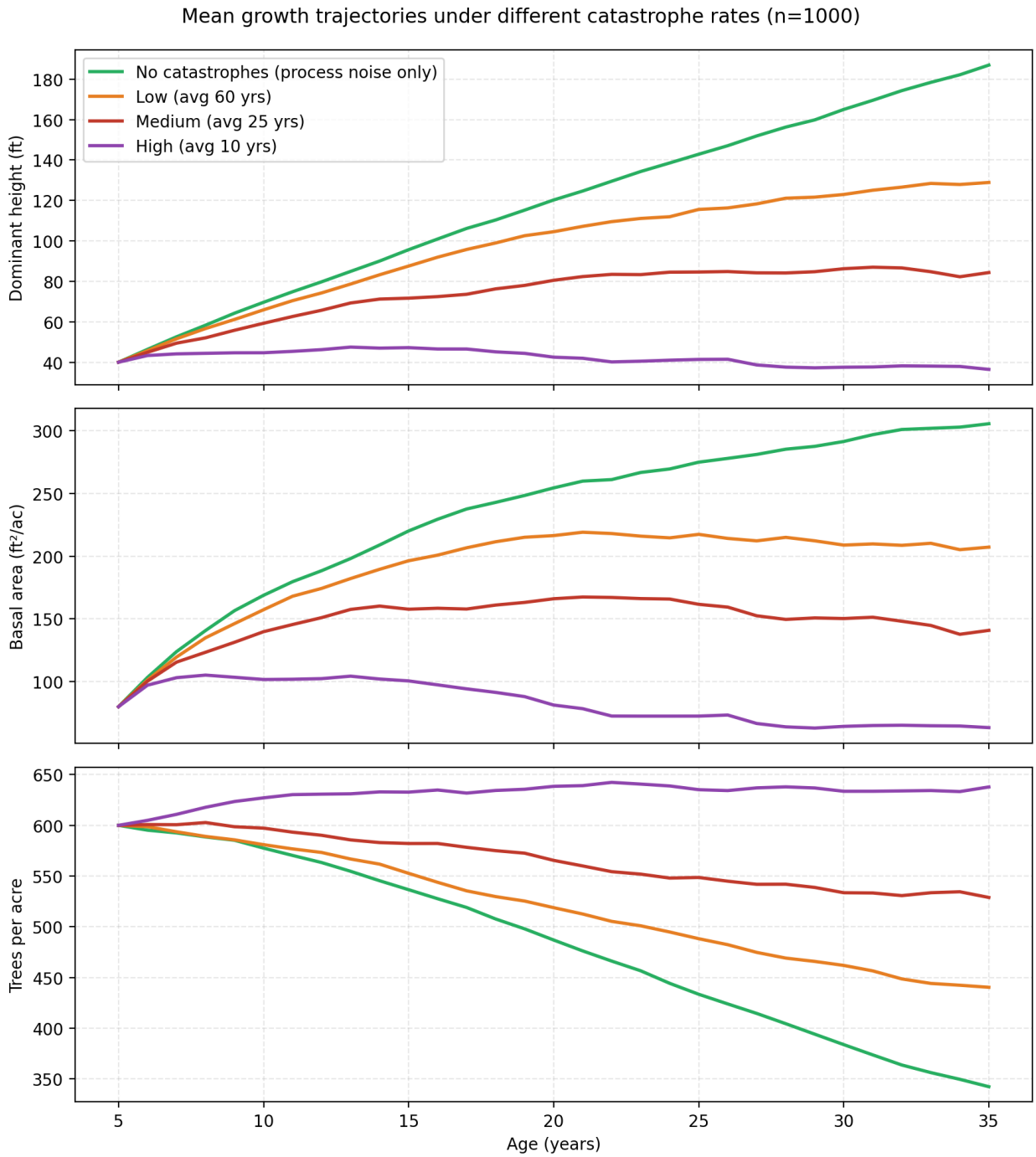

Figure 4.1: Stochastic growth by catastrophe probability.

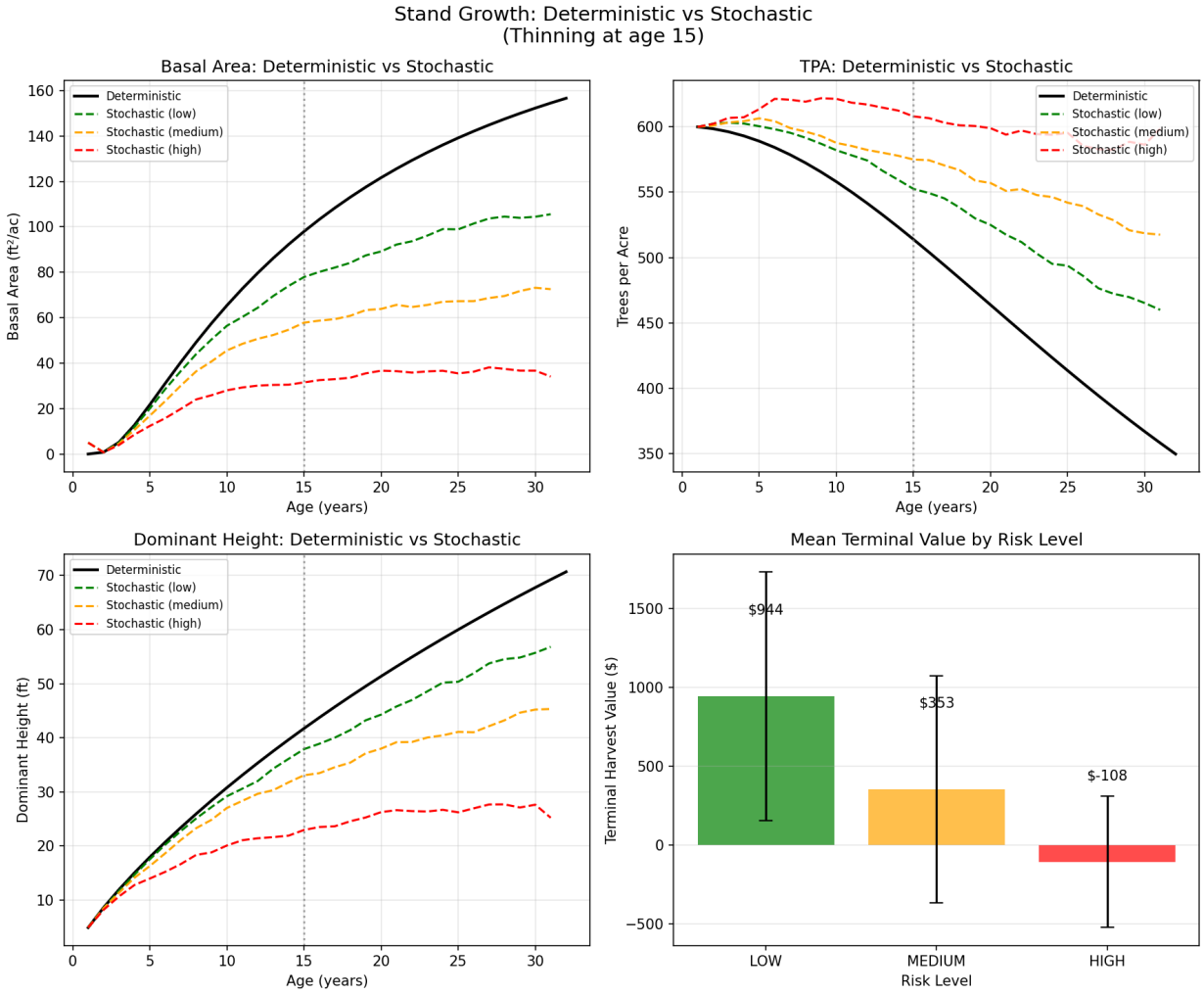

Figure 4.2: Deterministic vs. stochastic MDP comparison.

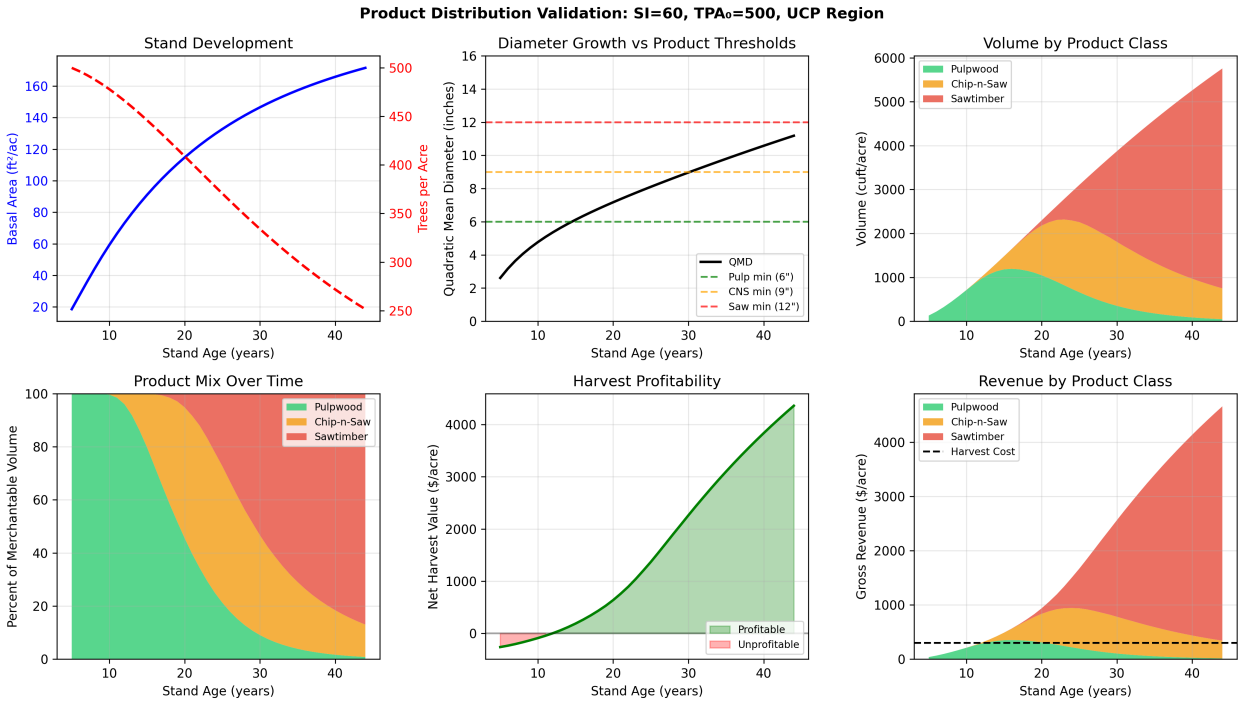

Figure 4.3: Product distribution following extended rotation (40 years).

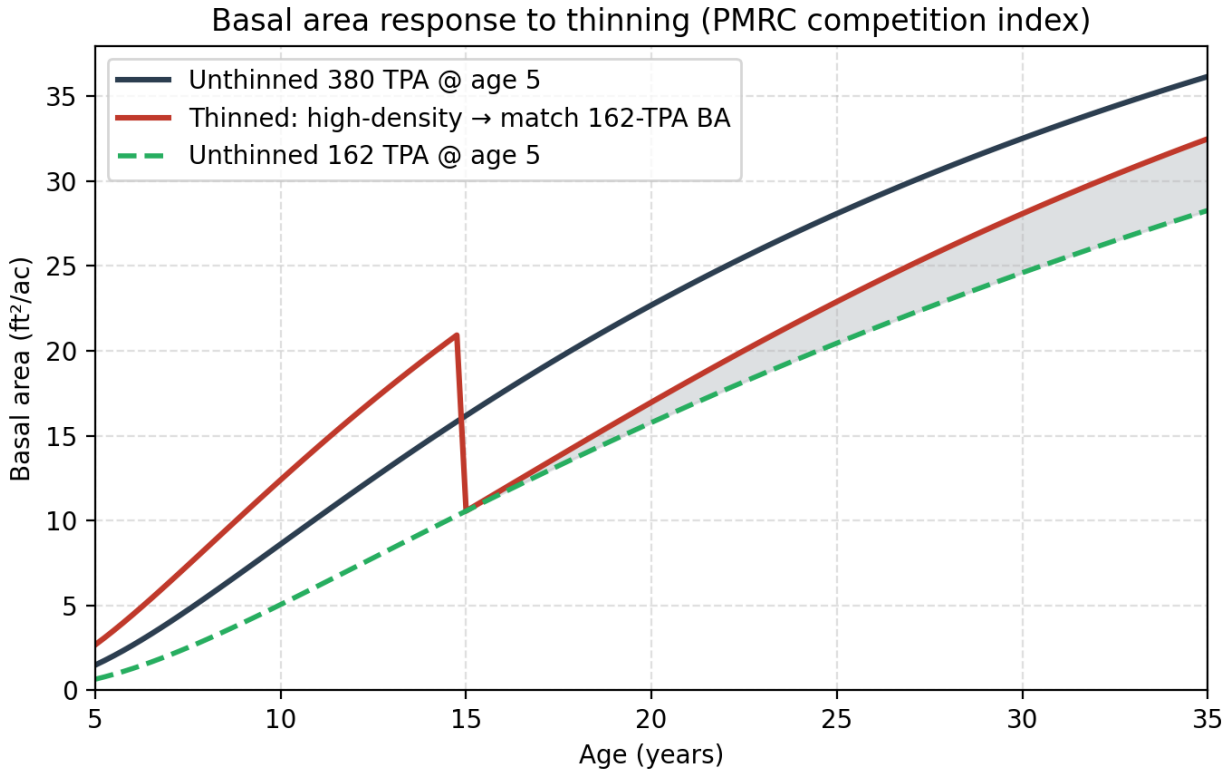

Figure 4.4: Thin response defined by the baseline growth model.

Chapter 5

Discussion

- Diminishing terminal value unsurprising given frequency of events, risk collapses stand quality and value - When disturbances are negligible, fixed rotations of 25-30 years are strong benchmarks. - Creating a stochastic wrapper on a deterministic growth model is ineffective.

Chapter 6

Conclusion

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Appendix A

Appendix Title